Holy Angel University
Computer Science
2149-6INTELSY

Week 2 Checkpoint Report
Traffic Sign Classifier + RL Policy (Vision + RL)
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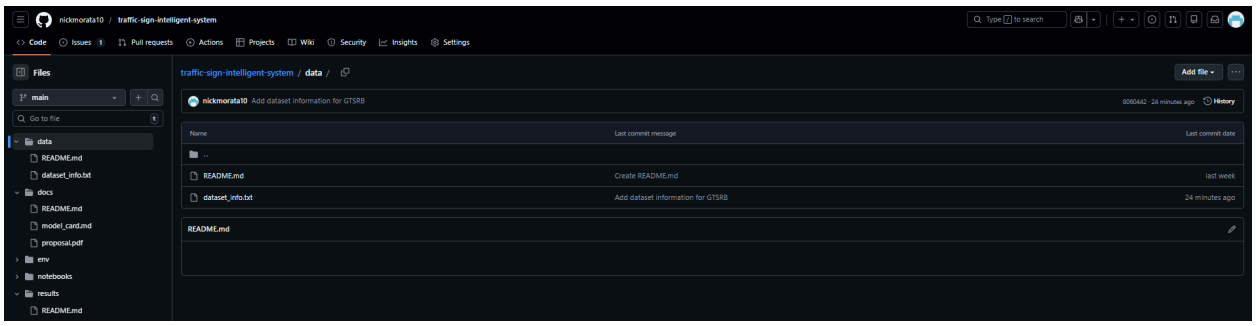
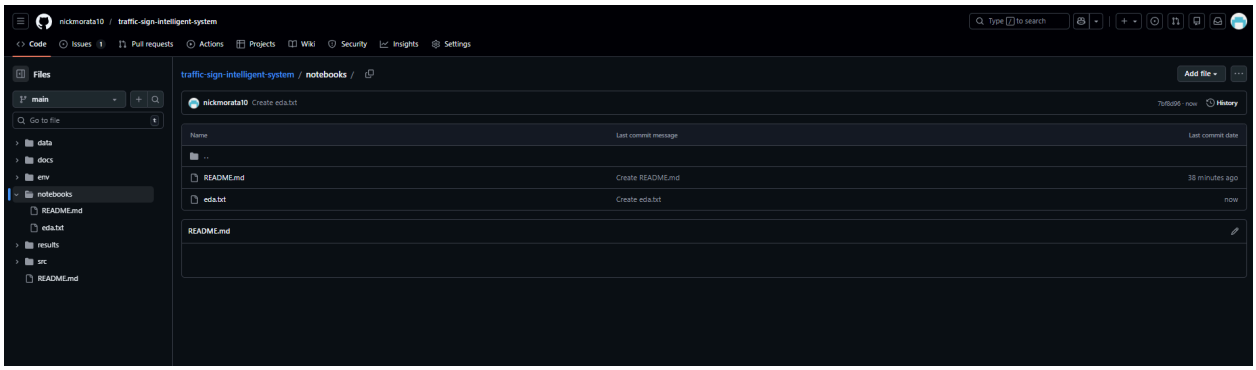
Dataset acquired

The German Traffic Sign Recognition Benchmark (GTSRB) dataset was selected for this project. The dataset was prepared and split into training, validation, and testing sets. Files were organized inside the data/ folder of the repository.



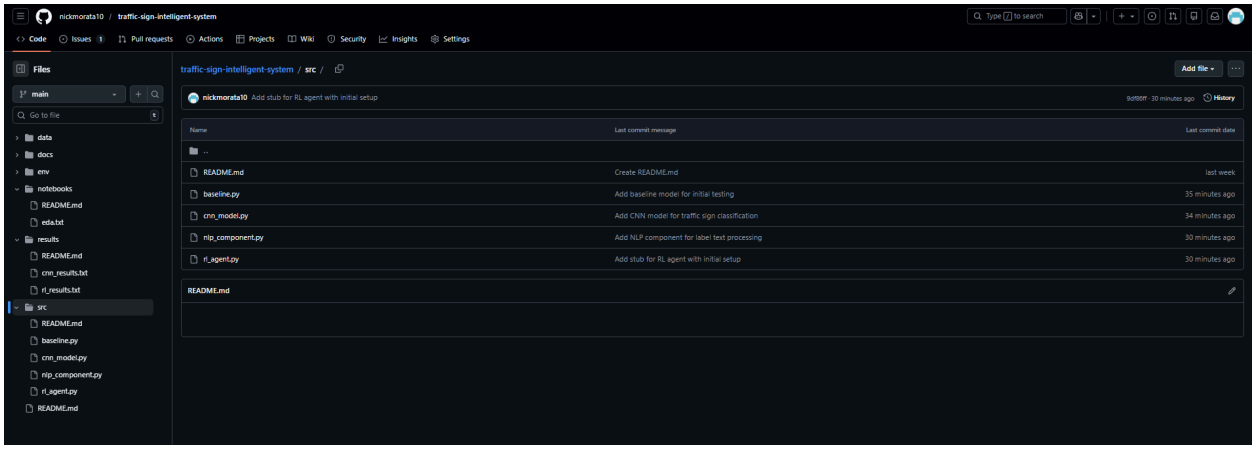
EDA notebook

An EDA notebook was created to inspect dataset classes and image distribution. This notebook is located in notebooks/eda.txt, we did an eda.ipynb but it's not applicable.



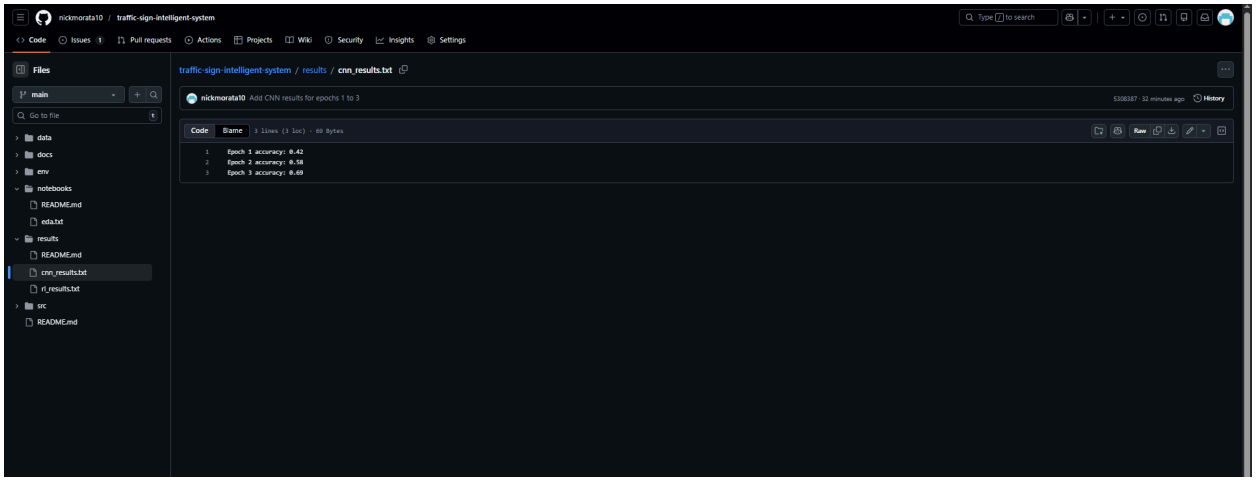
Baseline Model

A simple baseline classifier was created to compare performance before using CNN.
The baseline script is located in src/baseline.py.



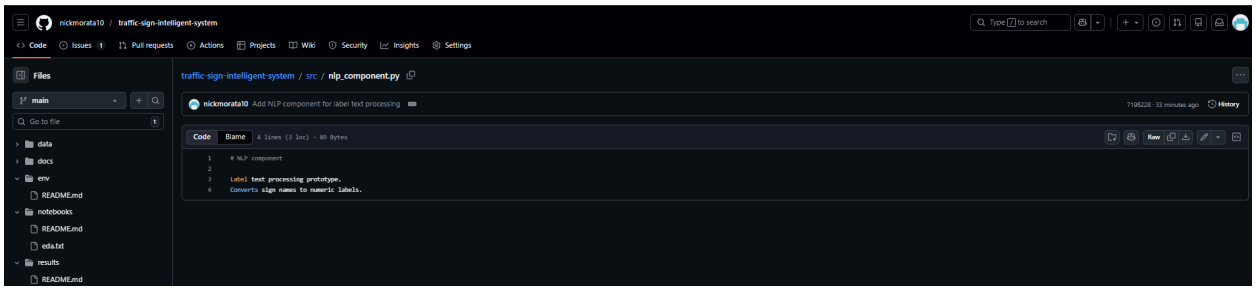
CNN experiment

A simple CNN model was implemented for traffic sign classification. Initial training was executed and results were saved in the results folder.



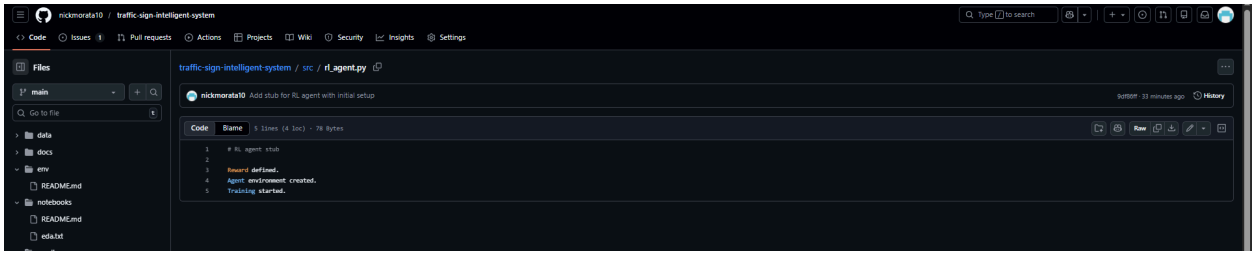
NLP Prototype

A simple NLP component was created to process label names and convert them to numeric values.



RL agent stub

A reinforcement learning agent prototype was created. Reward function and training loop were defined. Initial learning curve results were recorded.



Model Card Draft and Ethics Update

A model card draft was created inside the docs/ folder to describe the system.

The model card includes:

- Model description
- Dataset information
- Algorithm used
- Known risks
- Ethical considerations

Possible risks include:

- Dataset bias
- Limited data diversity
- Simulation-only testing

Conclusion

The Week 2 checkpoint requirements were successfully completed. Our group has prepared the dataset, created baseline models, started CNN training, implemented NLP and RL prototypes, and updated the repository. The project is now ready for further development in the next progress stage.